

Inert Parts Carousel

Admit, home, position, and audit one paper-bin request

E0 SYNTHETIC REPLAY — NO READER, STORAGE, MOTOR, SERVO, OR DISPENSING

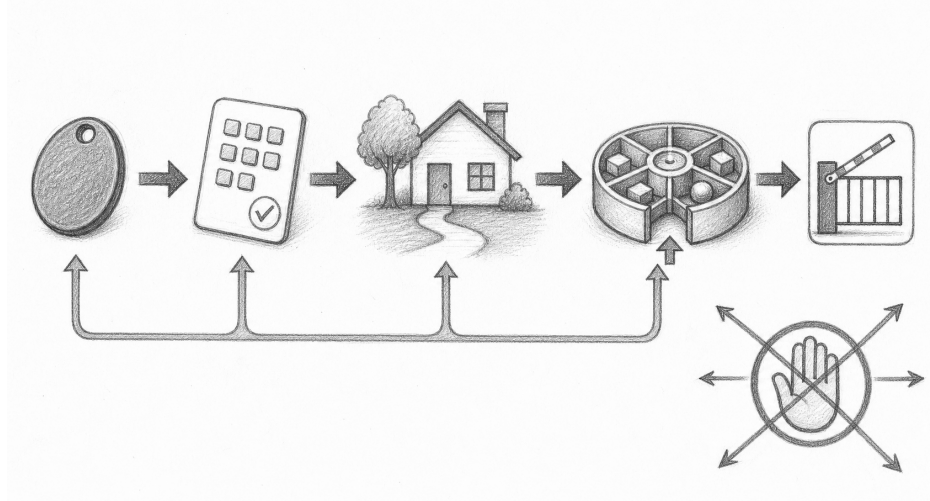
Lesson 051 — project
Prerequisites Lessons 047, 049, 050
Evidence result cells and fixture mirrors

Time 150–210 minutes
Board compile-only Mega replay
Position volatile logical coordinate only

Project boundary. `InertPartsCarousel` coordinates copied synthetic identity, key, home, stop, and presentation evidence with borrowed identity and homing policies, one privately owned logical sequencer, semantic gate intent, and caller-owned audit bytes. It owns zero pins, timers, interrupts, buses, endpoints, supplies, storage transports, or actuators.

1 Name the complete authorization invariant

The project does not treat a local identifier as a credential or a person. A known paper token selects a configured bin; an exact configured decimal confirmation grants a short-lived project authorization. Even that authorization cannot produce open-gate intent until every term below is true.



Alternative text: a pencil mnemonic links a round token, a keypad with a check mark, a house, a divided carousel, and a barrier-gate icon. A return line points back to the first four icons, while a crossed hand with outward arrows appears below the carousel. The drawing contains no labels, wiring, or powered hardware.

Required term	Exact E0 meaning
known identity	one binding in the committed caller-supplied image
matching confirmation	configured 1–4 digits, including leading zeros
durable admission	exact start record externally acknowledged from supplied bytes
successful home	fresh nonzero home epoch in this replay session
target reached	known logical coordinate equals the selected bin coordinate
healthy frame	current copied sources, valid time/skew, and inactive stop

The E0 result is a memory intent. It cannot authenticate an RFID card, prove a keypad press, establish physical home, move a carousel, open a gate, retain data through power loss, or dispense a part.

2 Read one transactional frame

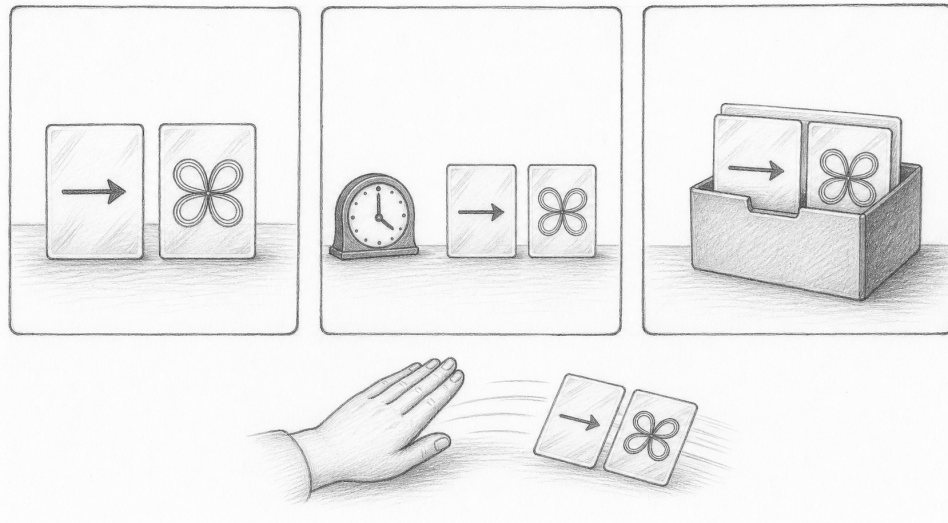
Each accepted call admits one timestamped frame and commits at most one bounded stage. Canonically absent identity and key payloads are fully zeroed; their named synthetic source and status remain valid. A present key batch has at most four qualified decimal digits and a zero tail.

1. Independently validate stop before unrelated frame fields.
2. Preflight identity, key, home, motion, presentation, and audit work.
3. Reserve adjacent start and terminal audit slots before authorization.
4. Export and acknowledge the exact start record before any motion intent.
5. Home and position through at most one logical transition per frame.
6. Publish bounded gate intent only after the full invariant holds.

7. Restore closed and all-off intent before terminal reconciliation.

Cancel dominates simultaneous confirm and digits. The collision records **ConfirmationConflict** and grants no authority. Repeated source sequences cannot add a digit. Authorization expires at its exact configured boundary before a colliding confirmation.

3 Coordinate two policies without partial motion



Alternative text: three pencil panels use paired cards marked with an arrow and a four-loop symbol. The first places the cards side by side, the second adds a clock, and the third places both cards in one box. Below, a hand approaches a separate card pair. The drawing is a sequencing mnemonic; it does not label previews, commits, coils, or hardware state.

Lesson 050 supplies only a signed semantic step request. The project translates that request to one exact Lesson 047 step, verifies both preview identities and both **canCommit()** results, and commits each exactly once only after no fallible work remains. Failure of either preview mutates neither policy. Coil bits come only from the committed Lesson 047 snapshot.

The private sequencer uses native logical zero. Its checked signed bounds cover every bin, zero, and the full Lesson 050 release/search excursion. There is no offset, rebasing, inferred shaft coordinate, or persisted physical position.

Treat qualified home as a synchronization boundary

On a new qualified home edge, the project derives but does not commit the Lesson 050 preview. It directly stops the private sequencer only if live, resets it unconditionally, and verifies logical zero, inactive phase, and zero coil intent. Only then may Lesson 050 publish its new home epoch. A stale preview cannot survive the reset generation change.

If direct stop or the zero/inactive/off check fails, the project stays all off, latches **PositionFault**, leaves position unknown, and publishes no home epoch. A simultaneous independent stop takes the stop path only.

4 Let valid stop bypass unrelated faults

A structurally valid, qualified active stop is the deliberate safety exception. It is admitted before malformed identity, key, home, presentation, or audit fields can delay it.

Private sequencer	Stop action	Required result
live or coils nonzero	direct stop(now)	coils zero, gate closed
idle and coils zero	do not mutate sequencer	home policy accepts stop
direct stop fails	all-off reset fallback	fault latched, no ordinary work

Stop cancels new authority and publishes closed/off intent. It never erases or downgrades an already latched fault. A malformed stop cannot use this exception and causes complete-frame rejection. Software stop intent is not an emergency stop, a stationary mechanism, or independently removed actuator power.

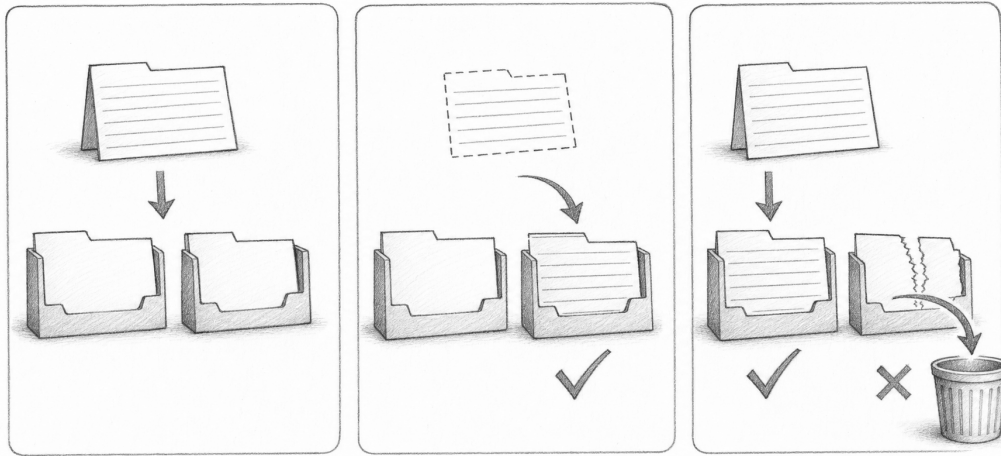
5 Open a semantic gate for bounded time

CarouselGateIntent::Open is a semantic result with an explicit expiry, not a servo pulse or physical gate state. At expiry the project first restores **Closed** and zero coil intent, then advances toward completion and terminal audit. No fault retries automatically.

No automatic dispensing. A token, confirmation code, logical home, or audit record never authorizes hazardous, valuable, access-controlled, unattended, launcher, or safety-device operation.

6 Reserve audit pairs before work begins

Each operation owns two adjacent 40-byte canonical records: one acknowledged authorization start and one terminal result. One remaining slot is **AuditFull**; the project never leaves a start in the final slot.



Alternative text: three pencil panels show a lined card above two adjacent file holders. The middle panel moves a dashed-outline card into the right holder and adds a check mark. The final panel keeps the left card, shows a torn card over the right holder, and points the torn pieces toward a wastebasket beside a cross mark. The drawing does not number slots or encode record contents.

Bytes	Offset	Canonical content
4	0	magic, schema version, encoded length
16	4	project, operation, authorization, sequence, occurrence time
4	20	kind, phase, bin, typed audit status
12	24	identity digest/revision/image generation, home epoch
4	36	two reserved zero bytes and CRC-16/CCITT-FALSE

The semantic C++ record is never persisted by its padded `sizeof`. The codec writes unsigned little-endian fields, zero reserved bytes, and a checksum over bytes 0–37. Valid capacities are exactly 2, 4, 6, or 8 records, with a 40-byte stride and a separate exact 40-byte candidate buffer.

7 Acknowledge exact external evidence

`previewAuditWrite()` and `previewAuditExport()` expose one retryable candidate. An external coordinator may write, synchronize, reread, and validate that candidate. `acknowledgeAuditWrite()` accepts only matching owner, generation, operation, slot, view metadata, checksum, bytes, synchronization, reread validation, and successful status.

A failed or indeterminate acknowledgement retains the candidate and blocks replacement, reset, and new input work until reconciliation. An exact repeated acknowledgement is idempotent; a changed repeat is an audit fault. These rules prove deterministic supplied-image semantics, not atomic media writes, retention, wear, or power-loss survival.

8 Reconstruct without inventing history

Recovery scans no more than eight slots and accepts only one canonical prefix. A start with an erased reserved terminal becomes inhibited recovery work for that same slot, operation, authorization, and identity binding. Recovery may write only **Recovered-Interrupted**; it cannot invent a new operation ID, authorization epoch, home epoch, bin, or physical position.

Reject unsupported schema, wrong project identity, corrupt checksum, noncanonical tail, terminal without start, mismatched pair fields, duplicate terminal, invalid sequence delta, a start in the final slot, or a non-erased mismatching reserved terminal. Restart begins gate closed, motion off, and position unknown even when every audit byte is valid.

9 Run the deterministic E0 workbook

1. Start from two canonical identity images, empty audit bytes, unknown position, closed gate, and zero coil intent.
2. Present one known copied token. Predict its exact bin, binding revision, and identity-image generation; then try unknown and repeated tokens.
3. Enter the exact configured code, including a leading-zero case. Collide cancel with confirm and verify conflict without authorization.
4. Export the start audit. Try foreign, changed, failed, and exact repeated acknowledgements before accepting the byte-identical record.
5. Replay release-first homing. At the qualified home edge, predict private sequencer stop/reset, logical zero, all-off intent, then a new home epoch.
6. Advance to the selected bin one due logical step per frame. Verify the coil mirror and that no gate-open intent precedes the complete invariant.
7. Observe bounded semantic open intent, exact expiry, closed/off intent, and the matching terminal audit candidate.
8. Collide valid stop with identity, confirmation, home, due step, arrival, gate expiry, audit-full, and presentation faults. Verify stop precedence.
9. Restart with a start-only prefix. Predict inhibited, unknown-position recovery into only the reserved terminal slot.
10. Inject corrupt pair fields, checksum, sequence ambiguity, full capacity, source fault, excessive skew, future time, rollover, shutdown, and a second byte-identical instance.

E0 intent result cells

frame	phase	bin	known?	logical position	home epoch
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
operation	authorized?	coil bits	gate intent	audit pending	fault
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____

10 Diagnose by precedence, not symptoms

Observed memory result	First question	Interpretation boundary
AuditFull	are two adjacent slots free?	no partial authorization
PositionFault	did stop/reset/zero-off pass?	no physical diagnosis
ConfirmationConflict	cancel, confirm, overflow, repeat?	no credential claim
AuditIndeterminate	do reread bytes identify the candidate?	no durability claim
PresentationFault	did primary phase remain intact?	display is not authority

11 Interpret only what E0 observes

Acquisition prediction. Canonical caller-owned buffers, valid child configuration, and current copied sources allow initialization and replay admission without claiming a hardware resource.

Acquisition observation. Host tests inspect lifecycle status, typed dispositions, audit candidates, exact byte views, operation identity, and complete snapshots in memory.

Acquisition interpretation. Success proves bounded policy composition over copied data. It does not prove an RFID reader, keypad, home sensor, display, storage medium, driver, motor, servo, or board was acquired.

Safe-state prediction. Construction, initialization, valid stop, fault, interrupted recovery, shutdown, and restart publish closed gate and zero coil intent; uncertain displacement leaves logical position unknown.

Safe-state observation. Inspect result cells, the private sequencer mirror, terminal attribution, and supplied audit bytes.

Safe-state interpretation. All-off memory intent proves software precedence only. It does not prove high impedance, de-energized windings, stationary hardware, a closed gate, or removed actuator power.

12 Keep the maximum fixture inside its gates

The complete eight-binding/eight-bin fixture includes two 160-byte identity images, one 160-byte identity candidate, at most eight 40-byte audit slots, one 40-byte audit candidate, borrowed Lessons 049/050 objects, the private Lesson 047 child, replay cells, and the coordinator.

Reusable children must remain at or below 128 bytes. The project coordinator measures 380 bytes against its reviewed 320-byte target and 384-byte hard ceiling. The Mega sketch measures 26,014 bytes flash and 1,933 bytes static SRAM against ceilings of 28,672 and 2,048 bytes. The conservative live stack/ISR path is 742 bytes and the reviewed reserve is 768 bytes, leaving 5,491 bytes after globals and that rounded reserve. Exceeding a gate stops for design review; storage does not move to the heap.

Compile-only means compile-only. A successful Mega build proves that the public API and inert fixture fit the toolchain and recorded budgets. It does not execute a result cell or qualify a powered circuit.

13 Leave E1 input and storage acceptance blank

E1 remains open. E1 may qualify exact low-energy input, presentation, and nonvolatile-storage adapters while actuator power remains absent. Supplied byte-image replay is not media durability evidence.

Exact RFID/keypad/home/stop specimens	_____
Primary datasheets and revision markings	_____
Logic/supply levels, polarity, buses, and pin map	_____
Exact display, LED, and resistor observation path	_____
Exact nonvolatile medium and storage adapter	_____
Write, sync, reread, rollback, and fault evidence	_____
Resource acquisition evidence	_____
Electrical safe-state evidence	_____
Reviewer, date, acceptance record	_____

14 Leave E2 powered carousel acceptance blank

E2 remains open. Do not add actuator wiring or power until the exact restrained mechanism, motor/driver, servo, supplies, protection, stop, and independent actuator-power removal are reviewed together.

Exact motor, winding, driver, and clamp topology	_____
Exact servo, gate geometry, and travel envelope	_____
Electrically authoritative formal schematic	_____
Separate current-limited actuator supply/common	_____
Measured stepper, servo-stall, aggregate current	_____
Thermal limits and de-energized hold behavior	_____
Stable restraint, pinch guarding, lightweight load	_____
Measured home, position, gate, stall, and collision	_____
Independent stop and actuator-power removal	_____
Named-person physical acceptance result	_____
Reviewer, date, acceptance record	_____

Future E2 work is limited to a stable, restrained, lightweight tabletop demonstrator. It excludes loose or high-inertia loads, lifting, sharp features, stored energy, vehicle or door-lock roles, unattended dispensing, launcher control, and any safety function. Indicator light alone cannot prove home, position, gate closure, de-energization, or independent power removal.

15 Software evidence and further work

Deterministic tests cover every bin and confirmation shape, full authorization invariant, both homing directions, joint preview, live/idle/failing stop, home synchronization, exact audit encoding and recovery, capacity edges, collision precedence, shutdown, restart, maximum composition, and two-instance fieldwise replay.

- Canonical compile-only Lesson 051 sketch.
- Deterministic project tests.
- Inert-parts-carousel interface.
- Lessons 049–051 design record.
- Deterministic testing contract.
- Safety model.

This untagged print edition has a searchable HTML alternative. It makes no PDF/UA, powered-input, storage-durability, motion, gate, dispensing, or physical-position claim.